

Impact of traffic sampling on LRD estimation

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Abstract. Network traffic sampling is an effective method for understanding the behavior and dynamics of a network, being essential to assist network planning and management. Tasks such as controlling Service Level Agreements or Quality of Service, as well as planning the capacity and the safety of a network can benefit from traffic sampling advantages. The main objective of this paper is focused on evaluating the impact of sampling network traffic on: (i) achieving a low-overhead estimation of the network state and (ii) assessing the statistical properties that sampled network traffic presents regarding the eventual persistence of Long-Range Dependence (LRD). For that, different Hurst parameter estimators have been used. Facing the impact of LRD on network congestion and traffic engineering, this work will help clarify the suitability of distinct sampling techniques in accurate network analysis.

Keywords: Traffic Sampling, Self-Similarity, Long Range Dependence

1 Introduction

In today's networks, due to the huge traffic volumes and speeds, it is increasingly harder and more expensive to understand the behaviour of a network through the analysis of the total number of packets. Consequently, the use of traffic sampling, which consists in electing a small number of packets for analysis, becomes essential. Even though sampling techniques reduce the amount of collected data drastically, sampling impacts on the performance of network equipment, and leads usually to a reduction in the estimation accuracy of network status and the statistical properties of traffic.

Since the identification of self-similarity and Long-Range Dependence (LRD) as properties of Internet traffic [9, 1, 11], representing a paradigm shift from Poisson and memory-less processes, there have been extensive research to estimate LRD in distinct network environments [10]. The presence of LRD in network traffic reveals the existence of statistically significant correlations across large time scales [7], meaning that traffic burstiness effects may be time-persistent.

Facing the impact of self-similarity and LRD on traffic engineering, namely on establishing proper network configurations and network congestion control, where bursts inside bursts may lead to unpredictable queuing delay and loss,

understanding the impact of sampling on LRD estimation becomes crucial. However, till present, no focus has been given to verifying if LRD property is maintained when traffic sampling is applied, and how sampling techniques affect the subsequent LRD estimation algorithms.

In this paper, the evaluation of the impact of selective analysis of network traffic will be explored at two levels: (i) in the overhead reduction when estimating network status; (ii) in the statistical properties of traffic, such as self-similarity and LRD, when comparing sampled to the original network traffic. In more detail, this involves: studying the different sampling techniques and the properties of network traffic; applying different sampling techniques to real-world datasets using the sampling framework proposed in [12]; and (iii) comparing the different traffic sampling techniques in terms of volume of data and accuracy in estimating the statistical properties of traffic. This will allow a better understanding of the behaviour and impact of sampling on network traffic analysis.

This paper is organized as follows: Section 2 provides background concepts and related work on traffic sampling and LRD; Section 3 details the methodology and tools used in the data processing and LRD estimation phase; Section 4 debates the evaluation results and main findings; Section 5 summarizes the conclusions and future work.

2 Background

This section provides a short overview of traffic sampling techniques and fractal properties of Internet traffic, such as self-similarity and LRD, and corresponding estimation methods.

2.1 Sampling

A way to lighten the burden of handling massive Internet traffic volumes is resorting to traffic sampling techniques to obtain a subset of the overall traffic for subsequent analysis. This subset should be yet representative of the whole traffic behaviour, allowing tasks such as traffic classification and protocol modeling, using substantially smaller data volumes. There are different sampling techniques and algorithms, with varying rates of data collected, computational cost and the ability of accurately representing the network behaviour. Due to its relevance, we briefly overview them next.

Sampling Techniques Overview - Sampling techniques can be broadly classified according to their: *Selection Scheme*, which determines the selection pattern of packets to be collected; *Selection Trigger*, which defines the type of trigger, which determines the start and the end of a sampling instance; and *Granularity*, which determines if sampling is carried out on a packet or flow basis (see Fig. 1).

As regards the selection scheme, the pattern to be collected can be ruled by a deterministic, random or adaptive function. While a probabilistic approach (random) can be used to overcome possible biasing of deterministic choice, adaptive

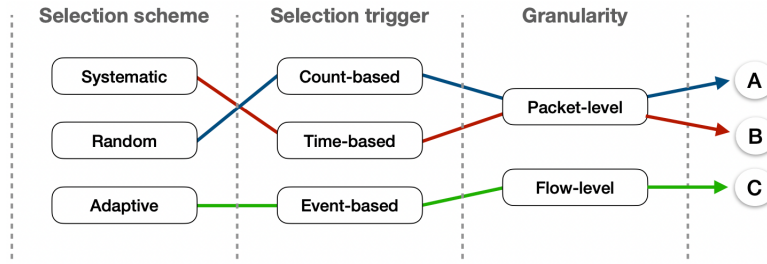


Fig. 1. Sampling techniques classification

approaches are able to adjust the pattern dynamically, e.g., based on ongoing traffic measurements. As regards the selection trigger, sampling can be prompted based on a counter, a timer or an event.

The sampling techniques within class A (see Fig. 1) are common in most sampling tools, e.g., Cisco NetFlow and sFlow. According to some authors, time-based triggers are less robust than count-based when applied to traffic characterization [4][12], being affected by the burstiness of network traffic. However, class B techniques may be suitable for applications requiring the analysis of consecutive packets, such as IDS [12] or for multi-channel wireless traffic characterization [13]. Event-based sampling techniques (class C), relying on content-dependent schemes, are oriented for monitoring a specific service or component, adapting the sampling frequency in response to some observed traffic event [5].

2.2 Properties of Internet Traffic

Self-similarity and scaling phenomena have dominated backbone Internet traffic analysis for the past two decades. However, self-similarity and LRD analysis is hindered by the difficulty of identifying dependence and estimating its parameters unambiguously [7].

Self-Similarity and Long Range Dependence - Internet traffic may exhibit intriguing temporal correlation properties and long memory [2], visible in plots of packet or byte traffic arrivals which are similar and bursty in different time scales. This behaviour is known as self-similarity or scale-invariance may hold true for different time granularity or scales, from a few hundred milliseconds up to hundreds of seconds. This means that a dilated portion of a traffic trace exhibits similar statistical characteristics than the whole. “Dilating” is applied on both amplitude and time axes of the sample path, according to a scaling parameter H called Hurst parameter. On the other hand, LRD is a long-memory property observed in large time scales.

A random process $X(t)$ (e.g., cumulative packet arrivals in time interval $[0, t]$) is said to be self-similar, with Hurst parameter $H > 0, H \in R$, if $X(t) =_d a^{-H} X(at)$, $\forall a > 0$, and $=_d$ denoting equality of all finite-order distributions [2].

The class of self-similar processes is usually restricted to that of self-similar with stationary increments (SSSI) processes, which are the “integral” of a stationary process [2]. The parameter H characterizes self-similar processes, but it is often used to label the LRD increments of SSSI processes. In this work, the expression “Hurst parameter of a LRD process” (characterized by γ) denotes actually the parameter $H = (\gamma + 1)/2$ of its integral SSSI parent process. The predominant way to quantify LRD is through the Hurst parameter, which is a scalar [7], and cannot be calculated definitively, only estimated. There are two categories of Hurst parameter estimators: those operating in the time domain, and those operating in frequency or wavelet domain.

2.3 Methods for estimating the parameters H and γ

Variance-Time Plot The variance-time plot method [2,11] studies the variance $\sigma^2(m)$ of the aggregated sequence obtained by dividing the time-series $\{x_j\}$ into non-overlapping blocks of length m and averaging them, i.e. $X_k^{(m)} = \frac{1}{m} \sum_{i=(k-1)m+1}^{km} x_i$ for $k=1,2,\dots,N/m$ [2]. The variance of the aggregated sequence can be estimated as the sample variance,

$$\sigma^2(m) = \frac{1}{N/m} \sum_{k=1}^{N/m} \left[x_k^{(m)} \right]^2 - \left[\frac{1}{N/m} \sum_{k=1}^{N/m} X_k^{(m)} \right]^2 \quad (1)$$

For large N/m and m , this variance obeys the power law $\sigma^2(m) \sim m^{\gamma-1} \sigma_x^2$. Thus, by linear regression on a log-log plot of $\sigma^2(m)$ as a function of m , the parameters γ and H can be estimated.

Rescaled Adjusted Range Statistic (R/S) The R/S statistic, one of the main time-domain methods for LRD estimation [14,6], is defined as:

$$\frac{R(n)}{S(n)} = \frac{1}{S(n)} \left\{ \max_{0 \leq l \leq n} \left[Y(l) - \frac{l}{n} Y(n) \right] - \min_{0 \leq l \leq n} \left[Y(l) - \frac{l}{n} Y(n) \right] \right\} \quad (2)$$

Assuming LRD of $\{x_j\}$ such as $S_Y(f) \sim c_2 |f|^{-r}$ for $f \rightarrow 0, 0 < r < 1$, when $n \rightarrow \infty$, its expected value is given by $E \left[\frac{R(n)}{S(n)} \right] \sim C_H n^H$, where C_H is a positive constant. To estimate the Hurst parameter using the statistic R/S, initially, the input sequence $\{x_j\}$ is divided in k blocks. Then, for each *lag-n*, R/S is computed at up to k starting points, taken evenly in different blocks. By linear regression on a log-log plot of $R(n)/S(n)$ as a function of n , the parameters γ and H can be estimated.

Periodogram This method plots the logarithm of the spectral density of the time series in comparison with the logarithm of the frequency. A function $I(\lambda)$ can be defined over a time series $X(t)$, as $I(\lambda) = \frac{1}{2\pi N} \left| \sum_{j=1}^N X_j e^{ij\lambda} \right|^2$, where λ is a frequency, N is the number of terms in the series, and X_j is the data under

analysis. As $I(\lambda)$ is an estimator of the spectral density, a series with LRD should have a periodogram which is proportional to $|\lambda|^{1-2H}$ close to the origin. Therefore, a regression of the logarithm of the periodogram on the logarithm of λ should give a coefficient of $1 - 2H$. The slope provides an estimate of H .

3 Experimental environment

This section describes the methodology and tools used to set up the testbed for LRD estimation over sampled datasets. For this purpose, both the sampling framework, the tools used to transform traffic traces into proper input data for the sampling framework and the LRD estimation algorithms are described next.

3.1 Sampling framework and datasets

The sampling framework used to obtain sampled data over large-scale public datasets was previously implemented and tested [12]. This framework includes a full range of sampling techniques as described in Section 2.1. The framework allows to apply these techniques to both online and offline traffic sampling in the form of pcap files. An integrated way of performing sampling, and the one used for obtaining sampled data sets, is using a Python script that applies all of the different sampling techniques previously referred to with pre-defined parameters, such as the sampling size and the interval between samples.

Regarding the network traffic used as input, this work resorts to public datasets from Optical Carrier OC48 and OC192 links (2.48Gbit/s and 9.95Gbit/s, respectively), gathered from an ISP infrastructure in the American west coast and maintained by CAIDA³.

3.2 Data preprocessing

CAIDA datasets undergone a preprocessing phase before serving as inputs both for the sampling framework and for LRD analysis. This preliminary phase involved the use of Mergecap tool for merging multiple CAIDA files, ordering data by timestamp, and Wireshark as traffic analyser.

Even though Wireshark has the possibility to turn a .pcap file into a .csv file with all data required for LRD estimation algorithms, namely the number of packets/bytes for each equally spaced time interval, it has limitations regarding the number of lines supported in the .csv file. To overcome this limitation, Editcap tool (with flag -i) was used to split the files composing the original dataset into smaller ones. In practice, Editcap reads part of or all captured packets from an *infile*, optionally converts them in various ways, and writes the resulting packet capture to an *outfile* (or outfiles), by default, in .pcapng format. Each outfile is created with an infix `_nnnnn[_YYYYmmddHHMMSS]`, comprising the serial number of the output file and the timestamp of its first packet.

³ Available at: <https://publicdata.caida.org/datasets/passive/>

The TShark protocol analyser was also used to generate statistics, such as the packets and/or number of bytes from the input file, and to collect packets/bytes statistics per time interval, with resolutions from seconds (s) to microseconds (μ s). The following code snippet shows how to obtain a specific statistic, the number of bytes per 0.1s interval, from all .pcap files in the current folder.

```
def tshark100ms():
    startdir='.'
    for root, dirs, files in os.walk(startdir):
        for file in files:
            if file.endswith('.pcap'):
                fName = os.path.join(root,file)
                packet = subprocess.run([f'tshark -r "{fName}"
                -qz io,stat,0.1 > {fName}0_1.txt'],
                capture_output=True,text=True,
                shell=True).stdout
                cmd = f"sed -i '/=====/,${!d}'
                {fName}0_1.txt"
```

Using the previous tools, it is possible to extract the number of packets/bytes per time interval from numerous text files, maintaining the chronological order in the resulting file. In particular, the granularity of 0.1s, 0.01s and 0.001s were used. Each file will serve as input for the statistics and LRD estimation algorithms. Instead of using the raw number of bytes, the data was normalized using the natural logarithmic function ($\ln$). The final text file containing the number of bytes per time interval was obtained after applying the Python *math.log* function. This file will serve later as input for the Selfis tool.

3.3 Selfis - LRD estimation

The evaluation results were obtained using the self-similarity analysis software tool SELFIS [8], capable of applying the Hurst estimation algorithms, described in Section 2.2, to an input file containing the time-series. Basic statistics, the graphical display of the power spectrum function and the autocorrelation function can also be obtained.

4 Results and discussion

The impact of traffic sampling on LRD estimation is assessed using two previously collected traces, a three hours long OC48 and two hours long OC192 publicly available, as described in Section 3.

LRD is comparatively analyzed resorting to three different time-domain Hurst parameter estimators, i.e., the *aggregated variance*, the *rescaled range* (R/S), and the *periodogram*. Each estimator is applied to three different time series corresponding to the amount of data from the two traces in the time windows of 0.1 sec, 0.01 sec and 0.001 sec. Notice the variation in the observation time window directly impacts on the communication overhead since smaller windows imply data being transmitted to the estimator more frequently. The same process is applied to the resulting traces from applying five sampling techniques, namely:

(i) SystC 1/100 - systematic count-based scheme with 1-out-of-100 packet selected for the sampled trace; (ii) RandC 1/100 - uniform probabilistic count-based scheme with 1-out-of-100 packet selected for the sampled trace; (iii) SystT 100/500 - systematic time-based scheme with the sampling frequency of 100ms-out-of-500ms; (iv) LP 100/200 - adaptive linear prediction scheme with a fixed sample size of 100ms and the initial interval between samples of 200ms; and (v) MuST 200/500 - multiadaptive scheme with the initial sample size of 200ms and the initial interval between samples of 500ms.

For the count-based schemes, the frequencies were selected based on [3], while the time-based frequencies were selected following the results in [12].

Considering that the main objective of using traffic sampling is to reduce the amount of data acquired and processed in multiple monitoring tasks, Figure 2 shows the data reduction achieved when considering the sampling schemes in use with different frequencies. Although the reduction ratio can be previously defined for count-based schemes, for time-based approaches, the bursty nature of the traffic leads to a variable volume of selected packets in the sample. Even though, regardless of the sampling strategy adopted, Figure 2 shows that the volume of data was reduced to $\approx 20\%$ (worst case) when compared with the original (i.e., unsampled) OC192 trace.

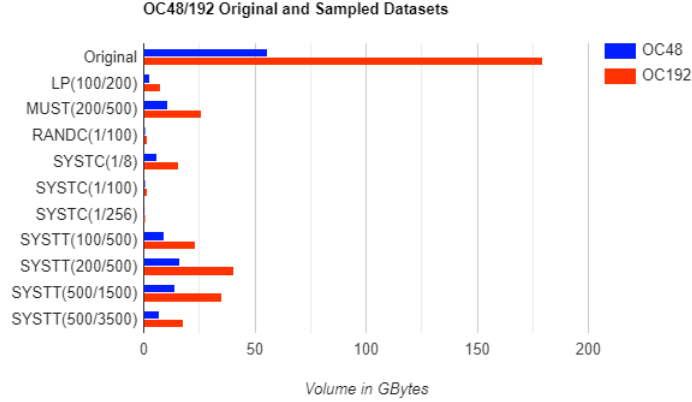


Fig. 2. Volume of data reduction through sampling.

Figure 3 illustrates the visual correlation of the number of packets (OC48) for two different time scales. Note that increasing the observation time-window does not lead to a smoother plot, which ratifies the self-similar behaviour described in Section 2.2. Although not represented here, a similar behaviour is observed for the OC192 trace.

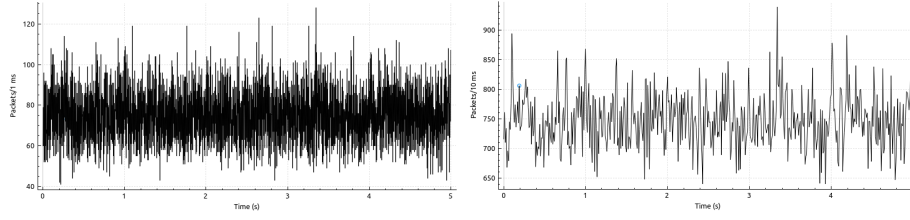


Fig. 3. Visual observation of self-similarity - total number of packets for a 0.001sec (left) and 0.01sec (right) time scale.

When comparing the sampling impact of estimating the Hurst parameter through the selected time-domain estimators, Tables 1-3 provide interesting outcomes. Firstly, for the aggregate variance (see Table 1), count-based techniques present a better relationship between the accuracy of LRD estimation and the amount of data required. However, as network measurements are typically multipurpose, the time-based multiadaptive scheme (i.e., MuST) may represent a good option. For instance, as discussed in [12], this technique provides better results for flow-based traffic classification.

Table 1. Hurst parameter using the aggregate variance estimator

	Trace OC48			Trace OC192		
Sampling scheme	0.1 sec	0.01 sec	0.001 sec	0.1 sec	0.01 sec	0.001 sec
Unsampled	0.857	0.923	0.908	0.784	0.828	0.835
SystC 1/100	0.815	0.835	0.727	0.832	0.818	0.804
RandC 1/100	0.814	0.837	0.740	0.793	0.819	0.804
SystT 100/500	0.429	0.611	0.776	0.822	0.720	0.714
LP 100/200	0.649	0.671	0.771	0.560	0.584	0.760
MuST 200/500	0.635	0.833	0.836	0.729	0.770	0.801

Figure 4 presents the visual representation of the R/S estimator for the unsampled OC48 trace aggregated bytes in 0.1 sec time-window. When applying this estimator to the sampled traffic (see Table 2), similar results for the aggregate variance are observed, including an accuracy degradation of the MuST technique for OC48 trace when aggregated over 0.1 sec time intervals. In this case, a comparable result is achieved for the OC192 trace for the same time aggregation. Here, the other adaptive sampling technique (LP) also provides valuable results for the OC48 trace, which ratifies the promising suitability of this type of sampling for multipurpose measurement tasks.

The comparative accuracy of time-based and count-based schemes is similar when resorting to the periodogram estimator. In this case, time-based techniques outperform count-based ones for particular time windows. In this sense, this estimator might be the most suitable to be combined with different measurement tasks while providing realistic estimations for LRD-based analysis.

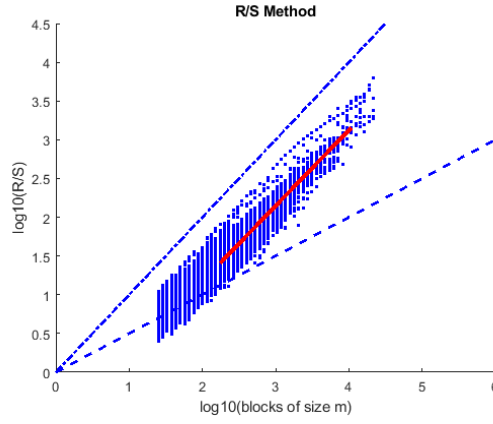


Fig. 4. Visual observation of R/S estimator for the OC48 trace.

Table 2. Hurst parameter using the R/S estimator

	Trace OC48			Trace OC192		
Sampling scheme	0.1 sec	0.01 sec	0.001 sec	0.1 sec	0.01 sec	0.001 sec
Unsampled	0.793	0.658	0.621	0.800	0.740	0.732
SystC 1/100	0.617	0.565	0.542	0.790	0.703	0.650
RandC 1/100	0.613	0.564	0.542	0.793	0.703	0.650
SystT 100/500	0.473	0.412	0.514	0.686	0.297	0.458
LP 100/200	0.747	0.409	0.585	0.598	0.474	0.658
MuST 200/500	0.281	0.614	0.626	0.445	0.650	0.774

Table 3. Hurst exponent for the periodogram estimator

	Trace OC48			Trace OC192		
Sampling scheme	0.1 sec	0.01 sec	0.001 sec	0.1 sec	0.01 sec	0.001 sec
Unsampled	1.053	0.663	0.589	0.857	0.676	0.755
SystC 1/100	0.731	0.543	0.505	0.742	0.658	0.648
RandC 1/100	0.822	0.539	0.507	0.867	0.661	0.649
SystT 100/500	0.540	0.371	0.514	1.035	0.435	0.652
LP 100/200	0.775	0.485	0.565	0.650	0.553	0.716
MuST 200/500	0.728	0.669	0.582	0.575	0.662	0.762

Although the 0.001sec observation time-window might increase the communication overhead related to measurement tasks, globally, this dimension presents a better accuracy regardless of the trace analysed. This conclusion is based on the sum of the estimation error for all compared sampling techniques.

5 Conclusions

The volume of data traversing modern networks demands that measurement-based tasks are able to provide accurate results without processing all the underlying packets. Traffic sampling is the primary strategy used to achieve this balance. However, different sampling techniques provide distinct accuracy results for different monitoring tasks. Thereby, this work presented a comparative

analysis of multiple sampling techniques applied to estimating temporal correlation and long memory behaviour of real backbone traffic. The results show that although varying in accuracy, sampling processes preserve the statistical traffic properties related to self-similarity while reducing the analyzed volume of data by 80%, in the worst case. Globally, higher accuracy was observed from count-based techniques, which also select the lower amount of data for samples. However, these results should be put into the context of multipurpose network measurements. For instance, if sampled traffic is also used for flow-based classification or intrusion detection, time-based techniques could provide better overall accuracy. Therefore, future work will consider different monitoring tasks and explore tuning adaptive techniques toward Hurst parameter estimation.

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